Twitter Sentiment Analysis Using Machine Learning and Transformers

Prepared by: Ashitha P | **Date:** 19 October 2026

1. Abstract

Sentiment analysis is a Natural Language Processing (NLP) technique for identifying opinions expressed in text. This project classifies tweets as positive, neutral, or negative using two approaches: a TF-IDF baseline and a fine-tuned RoBERTa model. Accuracy and Macro F1-score were used for evaluation. The best recorded result was 70.99% accuracy and 71.09% Macro F1-score using RoBERTa with random seed 42. Experiments also examined different seeds and class weighting.

2. Introduction

Social media contains large amounts of public opinion and customer feedback. Automatic sentiment classification can help analyse these opinions at scale. However, informal language, abbreviations, emojis, sarcasm, and ambiguous expressions make tweet classification challenging.

The objective was to compare a traditional machine-learning baseline with a transformer-based approach for three-class sentiment analysis.

3. Dataset

The project uses the sentiment-analysis subset of TweetEval, a benchmark for Twitter text classification. The three labels are:

- **Negative:** Dissatisfaction or criticism.
- **Neutral:** No clearly positive or negative sentiment.
- **Positive:** Satisfaction or favourable opinion.

The original dataset splits should be retained to support fair evaluation and avoid data leakage.

Dataset: TweetEval on Hugging Face.

4. Methodology

4.1 Data Preparation

The dataset was prepared for training and evaluation. For the baseline, tweets were converted into numerical features using Term Frequency–Inverse Document Frequency (TF-IDF). For RoBERTa, the appropriate tokenizer converted text into model inputs and the sentiment labels were mapped to three classes.

4.2 TF-IDF Baseline

TF-IDF represents text using word-importance scores based on their frequency within a document and across the dataset. These features were used in a traditional classification model to establish a baseline for comparison.

4.3 RoBERTa Model

RoBERTa is a transformer-based language model that captures contextual relationships between words. A pretrained RoBERTa model was fine-tuned on labelled examples for three-class sentiment classification.

4.4 Experimental Setup

Four experiments were compared:

1. TF-IDF baseline.
2. RoBERTa with seed 42.
3. RoBERTa with seed 7.
4. RoBERTa with class weighting.

Accuracy measures the proportion of correct predictions. **Macro F1-score** calculates the F1-score for each class and averages them equally, giving each sentiment class equal weight.

5. Results and Discussion

Table 1 presents the recorded experimental results.

Table 1: Model Performance Comparison

Model	Accuracy	Macro F1
TF-IDF baseline	58.94%	55.33%
RoBERTa (seed 42)	70.99%	71.09%
RoBERTa (seed 7)	70.61%	70.86%
RoBERTa (weighted)	69.12%	69.24%

Note: Verify all values against the final notebook outputs before submission.

5.1 Model Comparison

RoBERTa with seed 42 achieved the best recorded results. Compared with the TF-IDF baseline, accuracy improved by 12.05 percentage points and Macro F1-score by 15.76 percentage points. This suggests that contextual representations can improve sentiment classification compared with a traditional text-feature baseline.

5.2 Random Seeds and Class Weighting

The Macro F1-score differed by 0.23 percentage points between the two RoBERTa seed experiments. This indicates similar performance across these runs, although more repetitions are needed to establish reproducibility.

The class-weighted experiment achieved lower overall accuracy and Macro F1-score than the unweighted seed-42 experiment. Class weighting may help particular classes, but its effectiveness depends on class distribution and training configuration.

5.3 Class-Level Observations

The recorded neutral-class recall was 0.639, or 63.9%, if confirmed by the evaluation outputs. Neutral sentiment can be difficult to distinguish from positive and negative sentiment because of mixed opinions, indirect expressions, and limited context. A confusion matrix and per-class metrics can help identify common errors.

6. Challenges and Limitations

The main limitations include:

- **Ambiguity and sarcasm:** Literal wording may not reflect the intended sentiment.
- **Informal language:** Slang, emojis, abbreviations, and spelling variations complicate classification.
- **Class imbalance:** Unequal label distributions may affect performance across classes.
- **Experimental variability:** Results can change with random seeds and training configurations.
- **Generalisation:** Benchmark performance may not transfer directly to other domains or current social media data.

7. Conclusion

This project compared a TF-IDF baseline with a fine-tuned RoBERTa model for three-class Twitter sentiment analysis. RoBERTa with seed 42 achieved the best recorded performance, with 70.99% accuracy and 71.09% Macro F1-score. The results indicate that contextual language representations can outperform the traditional baseline. Class weighting did not improve the overall metrics in the recorded experiment.

Future work could include further seed experiments, confusion-matrix analysis, hyperparameter tuning, and evaluation on additional datasets. All reported results should be checked against the final experimental outputs.

8. Project Notebook

The implementation and experiments can be viewed in Google Colab:

Click here to view the project notebook.

9. References

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